# 外国語科目（数理・計算科学専攻）　　13 大修

## 英語

時間 午後２時〜午後３時

注意事項

1. 次の３問中より２問を選んで解答せよ．２問を越えて解答した場合は採点されない可能性がある．
2. 解答は１問ごとに別々の解答用紙に記入せよ．
3. 各解答用紙には必ず問題の番号および受験番号を記入せよ．

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問 1

次は、波の定義について書かれた文章である。全文を和訳せよ。

There appears to be no single precise definition of what exactly constitutes a wave. Various restrictive definitions can be given, but to cover the whole range of wave phenomena it seems preferable to be guided by the intuitive view that a wave is any recognizable signal that is transferred from one part of the medium to another with a recognizable velocity of propagation. The signal may be any feature of the disturbance, such as a maximum or an abrupt change in some quantity, provided that it can be clearly recognized and its location at any time can be determined. The signal may distort, change its magnitude, and change its velocity provided it is still recognizable. This may seem a little vague, but it turns out to be perfectly adequate and any attempt to be more precise appears to be too restrictive; different features are important in different types of wave.

G. B. Whitham, "LINEAR AND NONLINEAR WAVES"(1974) より

問 2
次の英文を読んで，下の (1)，(2) に日本語で答えよ．

While it is tempting to infer from the increasing volume of papers in our field that our community's collective creativity is on the rise, the cynics amongst us will be quick to point out that this phenomenon happens to coincide with the proliferation of word processing software and widespread computer availability. In short, it can be argued that the miracles of "cut-and-paste," "point-and-click," and "drag-and-drop" have facilitated the generation of papers through a process more mechanical and less thoughtful. As usual, the truth is probably somewhere in the middle. For some, efficient word processing provides an opportunity for more thinking time, since the paper-writing part is so much easier. For others, it is an opportunity to publish material that in the past they themselves would consider so marginal in scope that it would not justify the effort of writing an actual paper.

Whatever the case, the fact is that there are today, compared to 10 or 15 years ago, more specialized journals, more edited volumes, more monographs, and more voluminous conference proceedings than any of us can reasonably handle (fit in our bookcases, let alone read...). It is, therefore, not surprising that mere quantity is losing its meaning, as <u>we all seek to compensate for our inability to read all journals or attend all conferences by identifying the "best" or at least "most suitable" to our interests and quality criteria</u>.

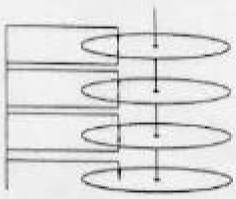
C.G. Cassandras, "Publish *and* Perish?" (1998) より

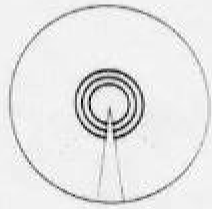
(1) ____部分において，"somewhere in the middle" が意味していることを説明し，著者がどんな状況を指してこのように主張しているのかを述べよ．

(2) ____部分を和訳せよ．

問3
次の英文はコンピュータの外部記憶装置である magnetic disk unit (磁気ディスク装置) の解説である．これを読んで後の (1),(2),(3) に日本語で答えよ．

A magnetic disk unit consists of one or more round plates with surfaces of magnetizable material, top and bottom. (中略) Each of the surfaces has its own reading head, and all the heads move in and out together, in order to read information at different radii from the center. (中略) The entire disk unit (except for the heads) rotates rapidly about the spindle at the center. (中略) The information on a disk is arranged in *tracks*, which are concentric circles. (中略) On each track, bits are represented magnetically, with 0 and 1 represented by opposite directions of polarity. (中略)


Schematic diagram of a disk unit.


Tracks on a disk surface.

In order to read a particular byte from the disk or to write a byte on the disk, we must do two things.

1. Position the heads so that a head is over the track of the desired byte.

2. Wait for the desired byte to rotate under the head as the disk spins.

Each of these operations might take an average of several hundredths of a second; (ア)<u>the actual time depends not only on characteristics of the disk unit, but on the number of tracks over which the heads must move and on the position of the byte relative to the heads at the moment the heads reach the proper track.</u> (中略) The next example suggests the time involved.

**Example.** Suppose we have a disk unit with the following characteristics:

1. The rotation rate is 3600 rpm, that is, 60 times a second.

2. Heads move in $5 + 0.01 \times t$ milliseconds (recall a millisecond is $10^{-3}$ second), if the distance moved is $t$ tracks.

3. Bits are packed into tracks at a density of 1000 bits per degree of arc, or 125 bytes per degree.

Further, suppose we wish to read 1000 consecutive bytes from a track that is 500 tracks away from the present position of the heads. Then the time to position the heads (called the *seek time*) is

$$5 + 500 \times 0.01 = 10 \text{ milliseconds}$$

The time to wait for the first byte to move under the head (called the *rotational latency*) can range from 0, if it is just at the right spot, to (イ)<u>16.7 milliseconds</u> if we have just missed it; the average will be 8.3 milliseconds, of course, for an average wait (*total latency*) of (ウ)<u>18.3 milliseconds</u>.

A. V. Aho and J. D. Ullman, "Foundations of Computer Science"(1992) より

(注)
bit ··· ビット (1桁の2進数(で表されるデータ)のこと.)
byte ··· バイト (コンピュータにおけるデータの基本単位. 1バイト＝8ビット.)

(1) 下線部(ア)を和訳せよ.

(2) 下線部(イ)の16.7ミリ秒とは何に要する時間であるか述べよ. またこの値は何を根拠にどのような計算で導かれたものかを述べよ.

(3) 下線部(ウ)の18.3ミリ秒とは何に要する時間であるか述べよ. またこの値は何を根拠にどのような計算で導かれたものかを述べよ.